

Francis Scott Key Bridge, Baltimore - Collapse (Mar 2024)

Civil Infrastructure & Construction · 2026-09-30

OVERALL CONDITION	8	8	0	0
CRITICAL	Findings	P1 Immediate	P2 Planned	P3 Monitor

Source footage	civil_key_bridge_collapse.webm
Footage length	451.3 s
Frames analysed	18 (one every 25.0 s)
Raw observations	9 (0 rejected as likely false alarms)
Detection model	Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)
Triage model	TypeSafe Jev (jev-1.13.0)
Processing time	25.2 s

How urgent is each priority

- P1

Immediate (within 24-72 h)
- P2

Planned (within 30 days)
- P3

Scheduled / monitor (next maintenance cycle)

1. Executive summary

NTSB drone footage from 26 March 2024, hours after the container ship Dali struck a main pier of the Francis Scott Key Bridge in Baltimore. The continuous steel truss spans have collapsed into the Patapsco River (F-02): a main pier is destroyed (F-03), a truss section is draped against another pier (F-03), and a whole span lies across the ship's bow, pinning it in place (F-04). Both approach roadways now end over open water (F-01, F-06). Containers at the bow are crushed (F-05), a containment boom along the hull suggests a possible fuel release (F-07), and the wreckage closes the shipping channel into the Port of Baltimore (F-08). This is a total structural failure; the report's job is to support safe closure, salvage and investigation, not repair.

Key risks

- Vehicles or people reaching the open ends of the approach spans (F-01, F-06).
- Unstable wreckage shifting during salvage onto crews and vessels (F-03, F-04).
- Hazardous cargo and fuel leaking from damaged containers and the ship (F-05, F-07).
- Port closed until the channel is cleared (F-08).

Findings by category

Structural failure	7
Other	1

2. Action plan

Immediate (24-72 hours)

- Hard road closures with barriers well back from both open ends (F-01, F-06).
- Marine exclusion zone and wreck marking in the channel (F-08).
- Dangerous-goods check of the damaged containers and fuel-leak response (F-05, F-07).
- Preserve evidence for the NTSB investigation.

Next 30 days

- Engineered salvage plan: cut and lift the truss off the ship and out of the channel (F-04, F-02).
- Open temporary alternate channels once surveyed (F-08).
- Assess the remaining piers and approach spans for damage (F-03).

Next 90 days

- Demolish or retain the approach spans based on the assessment; design a replacement crossing with pier protection sized for today's ships.
- Review pier protection (dolphins/fenders) on other bridges over busy shipping channels.

3. Findings register

All findings, most urgent first. Each is detailed with photos and repair steps in section 4.

ID	Time	Issue	Category	Severity	Priority	Action
F-01	00:25.0	Approach roadway ends abruptly at the collapse	structural failure	Critical (4.0)	P1	Shut down
F-02	00:25.0	Main truss spans collapsed into the channel	structural failure	Critical (4.0)	P1	Shut down
F-03	01:40.0	Truss section draped against a damaged pier	structural failure	Critical (4.0)	P1	Shut down
F-04	02:30.0	Truss span lying across the ship's bow	structural failure	Critical (4.0)	P1	Shut down
F-05	02:30.0	Containers crushed by falling bridge steel	structural failure	Critical (4.0)	P1	Shut down
F-06	05:00.0	Far-side approach span ends over open water	structural failure	Critical (4.0)	P1	Shut down
F-07	05:50.0	Containment boom along the hull (possible fuel release)	other	Critical (4.0)	P1	Shut down
F-08	04:10.0	Wreckage blocking the shipping channel	structural failure	Critical (3.9)	P1	Shut down

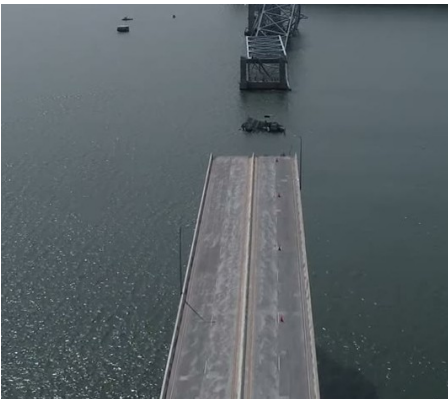
4. Detailed findings

F-01 Approach roadway ends abruptly at the collapse

P1 · Immediate (within 24-72 h)



Full frame with approximate location



Zoomed detail

Timestamp	00:25.0 (seen in 1 frame(s), from 00:25.0)
Component	bridge approach span (road deck)
Extent	Full road width
Category	structural failure (82% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	92% probability
Likelihood it is a real defect	97%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The approach road deck stops abruptly where the main spans have fallen. Anyone driving onto the bridge would drive straight into the gap.

Visual evidence: Road deck with lane markings ending in a clean cut over open water.

How to fix it

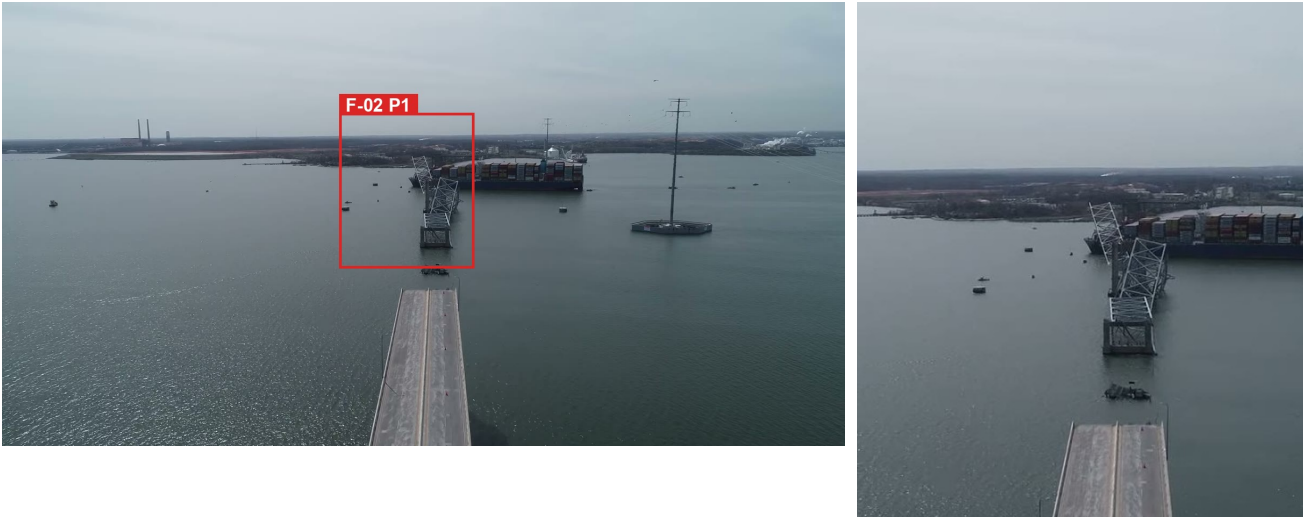
For this finding: Keep road closures at both ends with physical barriers well back from the gap; assess the remaining approach spans and piers.

Standard procedure

- 1 Close the structure and the area it could fall onto, including roads, waterways and paths below.
 - 2 Emergency engineering assessment: stability of what's still standing and risk of further collapse.
 - 3 Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
 - 4 Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).
- References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

F-02Main truss spans collapsed into the channel

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:25.0 (seen in 1 frame(s), from 00:25.0)
Component	continuous steel truss main spans
Extent	Main spans, hundreds of metres
Category	structural failure (100% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	92% probability
Likelihood it is a real defect	96%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The main steel truss spans have collapsed into the water, leaving only a pier and a piece of truss standing beyond the gap.

Visual evidence: Truss sections lying in the water and a single tower of steelwork near the ship.

How to fix it

For this finding: Keep the bridge closed and the waterway below restricted; engineers must assess the remaining piers and approach spans before any salvage work starts.

Standard procedure

- 1 Close the structure and the area it could fall onto, including roads, waterways and paths below.
- 2 Emergency engineering assessment: stability of what's still standing and risk of further collapse.
- 3 Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
- 4 Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).

References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

F-03 Truss section draped against a damaged pier

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	01:40.0 (seen in 2 frame(s), from 00:50.0)
Component	pier and collapsed truss
Extent	1 pier and one truss section
Category	structural failure (100% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	91% probability
Likelihood it is a real defect	97%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A large truss section has fallen against this pier and is tangled around it. The pier cap is damaged and the wreckage is unstable.

Visual evidence: Twisted steel truss leaning against a concrete pier column; bent and broken members.

How to fix it

For this finding: Keep the bridge closed and the waterway below restricted; engineers must assess the remaining piers and approach spans before any salvage work starts.

Standard procedure

- 1

Close the structure and the area it could fall onto, including roads, waterways and paths below.
- 2

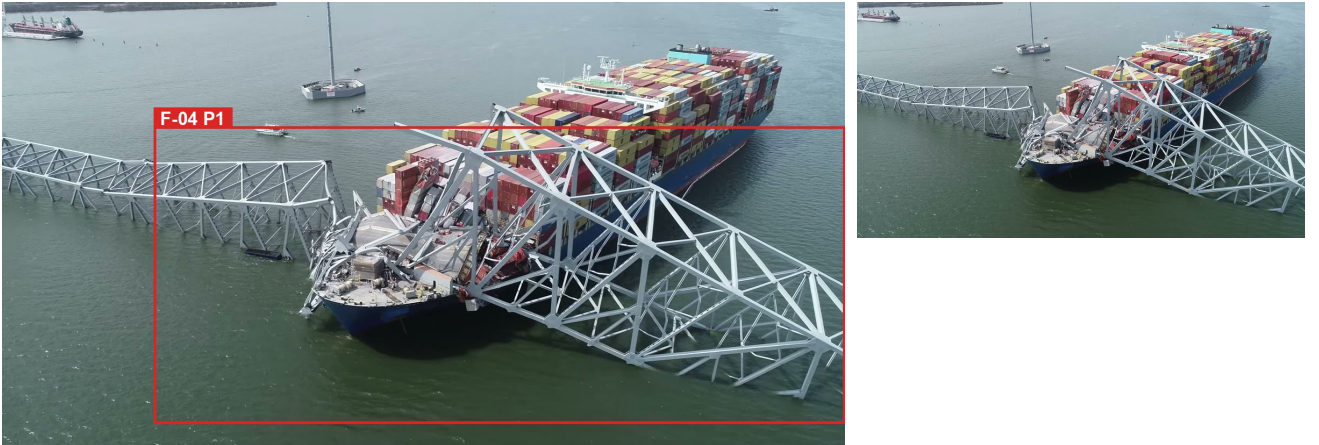
Emergency engineering assessment: stability of what's still standing and risk of further collapse.
- 3

Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
- 4

Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).
- References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

F-04 Truss span lying across the ship's bow

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	02:30.0 (seen in 1 frame(s), from 02:30.0)
Component	collapsed main truss span on the vessel
Extent	One truss span, ~100 m
Category	structural failure (100% confidence)
Severity	Critical, score 3.96 / 4
Safety / environmental hazard	81% probability
Likelihood it is a real defect	95%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A whole section of the steel truss has fallen across the ship's bow. The ship is pinned under the wreckage and can't be moved until it is cut away.

Visual evidence: Grey truss members spanning the bow, resting on the deck and containers, other truss sections in the water on both sides.

How to fix it

For this finding: Stabilise and cut the truss into sections with heavy-lift cranes under an engineered salvage plan before refloating the ship.

Standard procedure

- 1

Close the structure and the area it could fall onto, including roads, waterways and paths below.
- 2

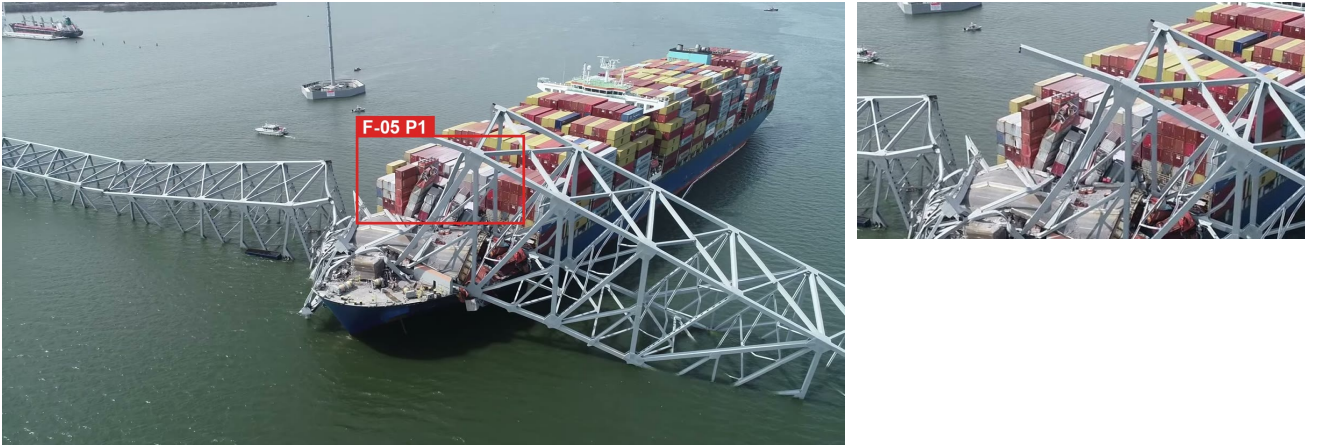
Emergency engineering assessment: stability of what's still standing and risk of further collapse.
- 3

Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
- 4

Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).
- References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

F-05 Containers crushed by falling bridge steel

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	02:30.0 (seen in 1 frame(s), from 02:30.0)
Component	containers on the ship's forward bays
Extent	Several containers
Category	structural failure (90% confidence)
Severity	Critical, score 3.98 / 4
Safety / environmental hazard	88% probability
Likelihood it is a real defect	95%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Containers at the front of the ship are crushed and knocked over by the falling steel. Some may hold hazardous goods and may be leaking.

Visual evidence: Crumpled and tilted containers under the truss at the bow.

How to fix it

For this finding: Check the cargo manifest for dangerous goods in the damaged bays; hazmat teams to inspect and monitor for leaks.

Standard procedure

- 1

Close the structure and the area it could fall onto, including roads, waterways and paths below.
- 2

Emergency engineering assessment: stability of what's still standing and risk of further collapse.
- 3

Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
- 4

Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).
- References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

F-06 Far-side approach span ends over open water

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	05:00.0 (seen in 1 frame(s), from 05:00.0)
Component	far-side approach span
Extent	Approach span
Category	structural failure (98% confidence)
Severity	Critical, score 3.97 / 4
Safety / environmental hazard	73% probability
Likelihood it is a real defect	93%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

On the far side the approach span also ends over open water.
Visual evidence: Road deck on piers ending at the gap, truss wreckage in the water next to it.

How to fix it

For this finding: Keep the far-side closure in place and assess the remaining piers.

Standard procedure

- 1 Close the structure and the area it could fall onto, including roads, waterways and paths below.
 - 2 Emergency engineering assessment: stability of what's still standing and risk of further collapse.
 - 3 Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
 - 4 Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).
- References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

F-07 Containment boom along the hull (possible fuel release)P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	05:50.0 (seen in 1 frame(s), from 05:50.0)
Component	water around the ship
Extent	Along the length of the hull
Category	other (70% confidence)
Severity	Critical, score 3.99 / 4
Safety / environmental hazard	74% probability
Likelihood it is a real defect	78%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Civil engineer
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

A yellow containment boom has been laid in the water along the ship's hull, which suggests fuel or oil may be leaking. The sheen itself can't be seen at this distance.
Visual evidence: Continuous yellow floating boom following the hull toward the bow.

How to fix it

- For this finding:** Keep booms in place and monitored; sample the water; check the ship's fuel tanks for damage.
- Standard procedure**
- 1 Have a qualified engineer review the frame and decide on an inspection or repair.

F-08 Wreckage blocking the shipping channel

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	04:10.0 (seen in 1 frame(s), from 04:10.0)
Component	navigation channel under the bridge
Extent	Channel width
Category	structural failure (100% confidence)
Severity	Critical, score 3.90 / 4
Safety / environmental hazard	62% probability
Likelihood it is a real defect	93%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Structural engineer + owner agency + emergency services
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Collapsed truss spans lie across the channel on both sides of the ship, closing the shipping channel into the port.
Visual evidence: Truss sections sticking up out of the water across the channel width.

How to fix it

For this finding: Keep the channel closed; mark wreckage with buoys; open temporary alternate channels once surveyed; plan wreck removal.

Standard procedure

- 1 Close the structure and the area it could fall onto, including roads, waterways and paths below.
- 2 Emergency engineering assessment: stability of what's still standing and risk of further collapse.
- 3 Stabilise, prop or remove unstable parts under engineer direction; plan detours and navigation closures.
- 4 Preserve evidence for the investigation; notify the regulator (FHWA/NTSB/state DOT for bridges).

References: 23 CFR 650 Subpart C (National Bridge Inspection Standards); AASHTO Manual for Bridge Evaluation

5. Methodology and limitations

How this report was produced

- Frames were sampled from the footage (one every 25.0 s, 18 in total).
- Detection: each frame was examined by a vision-language model (Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)) that listed every visible defect with its location, extent and visual evidence. It did not assign severity.
- Triage: each observation was scored by TypeSafe Jev (jev-1.13.0), a System One model that returns typed answers with probabilities: defect category, severity on a 5-level rubric, first action, the probability that the defect is genuine, and the probability of an immediate safety or environmental hazard.
- Rules applied by code: observations with less than 35% probability of being genuine were dropped; P1 if expected severity is 3.5 or higher or hazard probability is 60% or higher, or the recommended action is shutdown and severity is 2.5 or higher; P2 if severity is 1.75 or higher; otherwise P3. Findings with less than 60% probability of being genuine are capped at P2 and marked 'verify on site'.
- The same defect seen in neighbouring frames was merged into one finding.
- Repair procedures come from a reviewed playbook for this asset type; the per-finding fix is model-generated.

Severity scale

- 0: Cosmetic: no functional impact, note for records only.
- 1: Minor: early-stage wear; no loss of function; fix at next scheduled maintenance.
- 2: Moderate: measurable degradation or loss of output; fix within 30-90 days.
- 3: Major: active damage or significant loss of function; fix within 7-30 days.
- 4: Critical: active leak, fire, structural failure risk or danger to people or environment; act immediately.

Limitations

- Only what is visible from the air is assessed. Internal corrosion, wall thickness, electrical faults and gas leaks without a visible sign cannot be detected from RGB footage.
- Hotspot findings need radiometric thermal footage; on RGB footage they are not reliable.
- Locations are approximate boxes on the frame, not GPS coordinates.
- AI findings can be wrong. Every P1 and P2 finding should be confirmed on site before work starts.

Image credits

- Footage: "NTSB B-Roll - Aerial Imagery of Francis Scott Key Bridge and Cargo Ship Dali", National Transportation Safety Board, 26 March 2024, via Wikimedia Commons (public domain).